

ABDELRAHMAN GAMAL ZAYED

AI / Machine Learning Intern • Artificial Intelligence Engineering Student

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PROFESSIONAL SUMMARY

Artificial Intelligence Engineering student specializing in machine learning, computer vision, and Natural Language Processing (NLP). Hands-on experience developing end-to-end intelligent solutions using Python, Scikit-learn, TensorFlow, and PyTorch. Competent in executing data cleaning, exploratory data analysis (EDA), feature engineering, and robust model evaluation. Adept at translating complex engineering metrics into clear technical documentation and collaborating within cross-functional teams.

EDUCATION

Bachelor of Engineering in Artificial Intelligence Engineering

Oct 2023 – Jul 2028

New Ismailia National University • Ismailia, Egypt • **GPA: 3.65 / 4.00**

- **Relevant Coursework:** Machine Learning, Deep Learning, Data Science, Computer Science, Data Structures, and Object-Oriented Programming (OOP).

RESEARCH EXPERIENCE

Co-Author (Research Paper)

Under Review

Paper Title: Hybrid Metaheuristic Model for High-Fidelity Solar Power Forecasting

- Co-authored a peer-reviewed research paper proposing a hybrid framework that integrates an Extreme Learning Machine (ELM) with Improved Harris Hawks Optimization (IHO) for enhanced solar power prediction.
- Benchmarked predictive performance against CatBoost, XGBoost, Transformer architectures, and standalone ELM models using R^2 , RMSE, and MAPE validation metrics.

PROFESSIONAL EXPERIENCE

Machine Learning Intern

Jul 2025

National Telecommunication Institute (NTI) • Remote

- Engineered a predictive classification pipeline for diabetes risk assessment utilizing structured clinical health datasets.
- Executed end-to-end data preprocessing, target-driven feature engineering, and compared 4 supervised learning algorithms (Logistic Regression, Random Forest, SVM, and Decision Trees) utilizing accuracy, precision, recall, and F1-score.

AI & Machine Learning Trainee

Apr 2025 – Jun 2025

SprintUp Program by Sprints • Remote

- Developed a supervised learning workflow leveraging Python, Scikit-learn pipelines, structured EDA, and dynamic data visualization to transform raw datasets into model-ready features.

Deep Learning Trainee

Sep 2025

Egyptian Engineers Syndicate • Online

- Completed intensive technical training covering deep neural networks, transfer learning, regularization, and overfitting mitigation strategies while logging loss curves and experimentation architectures.

TECHNICAL PROJECTS

Diabetes Prediction Classification Model

Machine Learning & Healthcare

- Built a healthcare risk classification pipeline with hyperparameter tuning and stratified cross-validation; optimized model configurations against class imbalance to minimize critical false negatives.

Multilingual Fake News Classification

Natural Language Processing

- Designed a bilingual text classification system optimized for Arabic and English text datasets utilizing custom text preprocessing, tokenization, cleaning, and statistical feature extraction.

Facial Emotion Recognition Model

Deep Learning & Computer Vision

- Developed a 7-class facial emotion image classifier leveraging custom CNNs, TensorFlow/Keras, real-time validation monitoring, data augmentation, and OpenCV pipelines to prevent overfitting.

TECHNICAL SKILLS

Programming & Tools: Python, C, C++, Jupyter Notebook, Git, GitHub, PyCharm, Technical Documentation

Machine Learning: Supervised Learning, Classification, Regression, Feature Selection, Feature Engineering, Tuning, EDA

Deep Learning & CV: TensorFlow, Keras, PyTorch, OpenCV, Convolutional Neural Networks (CNNs), Transfer Learning

Core Algorithms: Logistic Regression, Random Forest, SVM, Decision Trees, XGBoost, CatBoost, ELM, Transformers

Soft Skills & Langs: Analytical Problem-Solving, Team Collaboration, Time Management | Arabic (Native), English (Intermediate)

LEADERSHIP & VOLUNTEERING

AI Head • Microsoft Students Club • Suez Canal University (SCU)

Present

- Direct AI-focused student workshops, hackathons, and peer learning pathways across ML, computer vision, and NLP.

Volunteer • Egyptian Youth Entity (EYE)

Present

- Collaborate with organizing committees to drive youth-led community development initiatives.